

C A-vs-B Human Evaluation Interface

The Survey Interface is a web-based evaluation tool designed to allow participants to compare outputs from two models (A and B). For each evaluation item, users are presented with the same input prompt along with two model responses shown in random order. Participants are asked to choose the better response based on specific criteria—such as Recommendation Relevance and Response Naturalness—or select "Tie".

Previous Chat History

I am looking for some mellow songs to relax to.

Here's a relaxing track by a Singer/Songwriter.

Time by Canyon City

That was alright, but I want something a bit more upbeat.

Model A Prediction

Okay, how about this slightly more energetic track from the same artist?

Cardinal by Mt. Joy

Model B Prediction

Let's try a track with a more upbeat tempo in the Pop genre.

Trenches by Gabe Bondoc

Recommendation Relevance: Which model provided better music recommendations?

Please consider all previous history.

☐ Model A ☐ Model B ☐ Tie (Both equally good)

Response Naturalness: Which model provided better text response?

Please consider all previous history + Relationship with current turn's music recommendation.

☐ Model A ☐ Model B ☐ Tie (Both equally good)

Next Question

Figure 4: Survey Interface of Subjective Evaluation.

D Comparison Music Recommendation Models

We compare our LLM-based conversational recommender system, TALKPLAY, against five representative baselines that span different recommendation frameworks (sparse, similarity-based, generative retrieval) and item representation approaches (text, audio, listening history), each of which represents a distinct methodology for recommendation:

BM25 [40] represents a classical text-based sparse retrieval approach. It utilizes term frequency-inverse document frequency (TF-IDF) scoring to match user queries with text-based item representations. While computationally efficient and interpretable, this method relies solely on lexical matching and cannot capture semantic relationships or cross-modal patterns.

NV-Embeds-V2 [29] advances beyond sparse retrieval by implementing a text-based deep similarity retrieval framework. The model learns dense neural embeddings for both queries and music descriptions, enabling semantic matching that can capture latent relationships beyond exact word matches. This model represents a strong baseline as an LLM-based embedding extractor model that has demonstrated superior retrieval performance on the MTEB benchmark.

CLAP [49] extends the similarity-based paradigm to multiple modalities by jointly embedding audio and text. Through contrastive learning, it aligns the representations of music audio with corresponding textual descriptions, enabling cross-modal retrieval. This model represents the current state-of-the-art in audio-text music retrieval, providing a strong baseline for multimodal understanding.

SASRec [26] represents a sequential recommendation approach based on self-attention mechanisms. The model treats each music track as a unique token and learns to predict the next item in a sequence using transformer-based architecture. While effective for capturing sequential patterns in user listening history, it lacks the ability to process natural language queries or content information, making it primarily suitable for next-item prediction based on historical interactions. For the architecture in the default version of SASRec, we use two self-attention blocks with 50 latent dimensions. The model is trained using the Adam optimizer with a learning rate of 0.001, batch size of 128, dropout rate of 0.2, and maximum sequence length of 50.

TIGER [39] implements a generative retrieval framework that combines semantic understanding with residual vector quantization. Instead of using unique item IDs, it represents each music track using Semantic IDs composed of SentenceT5 embeddings and residual vector quantization (RVQ). The model is implemented using a transformer-based encoder-decoder architecture containing 4 layers each with 6 self-attention heads, using a vocabulary of 1024 semantic tokens (256 x 4). The model is trained with a batch size of 256, using a learning rate of 0.01.

For text-based content retrieval models (BM25, NV-Embeds-V2, TIGER), item representations include metadata and semantic tags,⁴ excluding lyrics, playlist context, and audio modalities. We initially included lyrics in our item representations but found that this led to significant performance degradation. Therefore, we excluded lyrics from the final item representations. CLAP, being an audio-text joint embedding model, restricts item representations to only the audio modality, while SASRec, as a sequential recommendation model, represents items using only random IDs without any content information.

⁴{Title} by {Artist} from {Album} {Semantic tags} {Year}